15 Tracey is going to choose a main course and a dessert in a cafe.
She can choose from 8 main courses and 7 desserts.

Tracey says that to work out the number of different ways of choosing a main course and a dessert you add 8 and 7

(a) Is Tracey correct?
You must give a reason for your answer.

No. Each of the 8 mains can go with any of the 7 desserts.

Multiply: 8 × 7 = 56 different meals.

1 mark
Use the product rule, with a reason

(1)

12 teams play in a competition.
Each team plays each other team exactly once.

(b) Work out the total number of games played.

Each of the 12 teams plays 11 other teams.

12 × 11 = 132 counts every game twice.

1 mark
Count all team-opponent pairings

Divide by 2: 132 ÷ 2 = 66 games.

1 mark
Count each game once

66

(2)

(Total for Question 15 is 3 marks)

16 Solve $(x-2)^2 = 3$

Give your solutions correct to 3 significant figures.

(Total for Question 16 is 2 marks)

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